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BS.IT 5TH Semester
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Topic 4: SGML

SGML

SGML stands for Standard generalized markup language is a Standard generalized markup language that makes use of a superset of extensively used markup languages like HTML and XML. It is used for marking up files and has the gain of now no longer depending on a particular application.

It is basically derived from **GML** (Generalized Markup Language), which allowed users to work on standardized formatting styles for electronic documents. It was developed and standardized by the **International Organization for Standards** (ISO) in 1986. SGML specifies the rules for tagging elements. These tags can then be interpreted to layout factors in specific ways.

It is used extensively to manipulate massive files which are a concern with common revisions and want to be published in one-of-a-kind formats due to the fact it's far a massive and complicated system, it isn't always but extensively used on private computers.

HTML was theoretically an example of an SGML-based language until **HTML 5,** which browsers cannot parse as SGML for compatibility reasons. The SGML is extended from GML and later on it is extended to HTML and XML.

Structure of SGML:

<mainObject>

<subObject>

</subObject>

</mainObject>

The extension of SGML files is:

File_Name.sgml

Syntax:

<NAME TYPE="user">
AREEJ
</NAME>

Example 1: In this example, we will write code in SGML

```
<EMAIL>
    <SENDER>
        <PERSON>
            <FIRSTNAME>GEEKS FOR GEEKS</ FIRSTNAME>
        </PERSON>
    </SENDER>
    <BODY>
        <p>A Computer Science Portal For Geeks</ p>
    </BODY>
</EMAIL>
```

OUTPUT:

GEEKS FOR GEEKS

A Computer Science Portal For Geeks

Example 2: In this example, we will see how to write code in SGML and its element.

```
<EMAIL>
    <RECEIVER>
        <PERSON>
            <FIRSTNAME>Krishna</FIRSTNAME>
        </PERSON>
    </RECEIVER>
    <BODY>
        <p>It is a name of the person.</ p>

    </BODY>
</EMAIL>
```

OUTPUT:

Krishna

It is a name of the person.

Components of SGML:

- SGML provides a way of describing the relationships between these entities, elements, and attributes, and tells the computer how it can recognize the component parts of a document and it is based on the concept of a document being composed of a series of entities (object).
- It provides rules that allow the computer to recognize where the various elements of a text entity start and end.
- ***Document Type Definition (DTD)*** in SGML is used to describe each element of the document in a form that the computer can understand.

SGML is the simplest medium to produce files that can be read by people and exchanged between machines and applications in a straightforward manner. It is easy to understand by the human as well as the machine.

Characteristics OF SGML

- The SGML Declarations.
- The Prologue, containing a DOCTYPE declaration with the various markup declarations that together make a DTD i.e Document Type Definition.
- The instance itself, containing one top-most element and its contents.

Components of an SGML Document :

There are mainly three components of SGML document. They are –

1. SGML Declaration
2. Prolog
3. Document instance.

Features of SGML

- Platform-independent
- Elements identified by various tags
- Generic types and attributes are found in elements.
- Use of delimiters and special characters on a regular basis

Drawbacks of SGML

- It has a difficult connecting process.
- Writing SGML code is quite difficult.

Advantages

- It has the capability to encode the full structure of the document and can support any media type.
- It is of much more use than HTML which provides capabilities to code visual representation and not to structure the real piece of information.
- Separates content from appearance.
- SGML files encoding is allowed for more complex formatting as compared to HTML.
- The Stylesheets present in SGML make the content to use for different purposes.
- Extremely flexible.
- Well supported with many tools available because of ISO standard.

Disadvantages

- It may be typical to code software in SGML.
- Tools that are used in SGML are expensive.
- It may not be used widely.
- Special software is required to run or to allow the document to display.
- Creating DTD’s requires exacting software engineering.